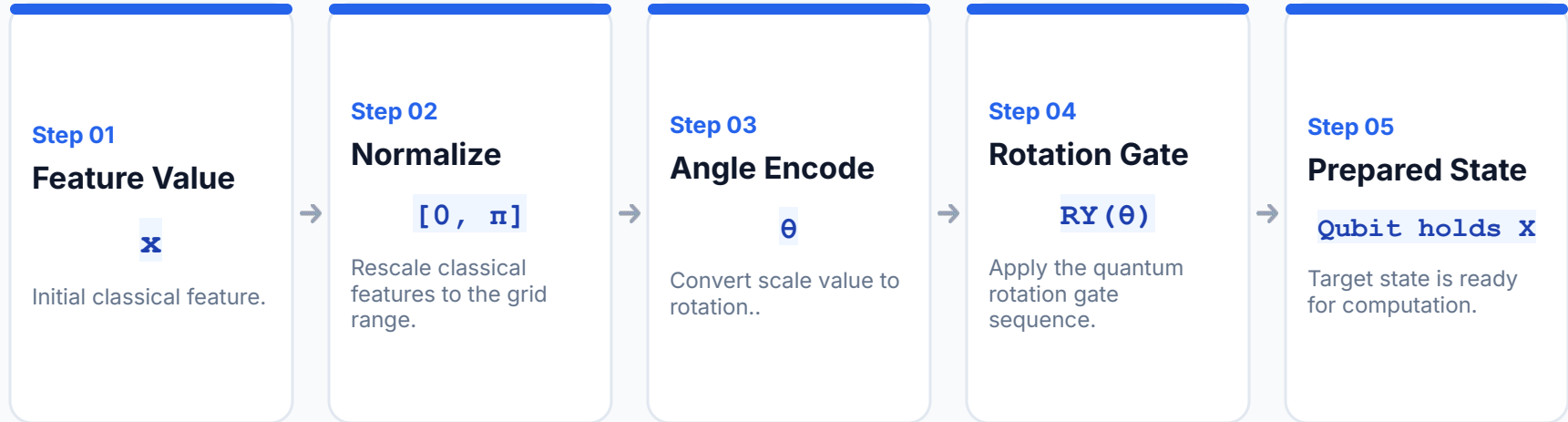

Quantum Encoding and Mapping TNBC Features

Presenter

Patrina Pun

1. Quantum Encoding Pipeline



3. Encoding Equations

Step 01

Normalize

```
x_norm= (feature - min) /  
(max - min)
```

Rescale classical features to the grid range.

Step 02

Angle Encode

```
 $\theta = \pi \cdot x\_norm$ 
```

Convert scaled values into quantum rotation angles.

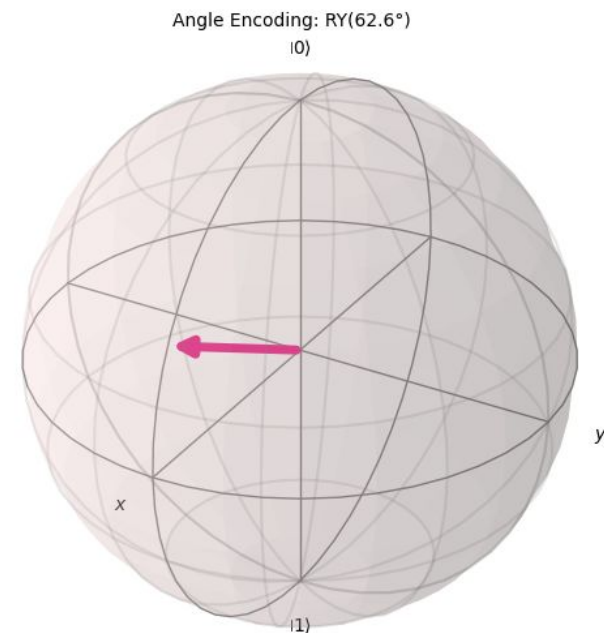
Step 03

Apply RY(θ)

```
 $R_Y(\theta) \mid \theta = \cos(\theta/2) \mid \theta \rangle + \sin(\theta/2) \mid 1 \rangle$ 
```

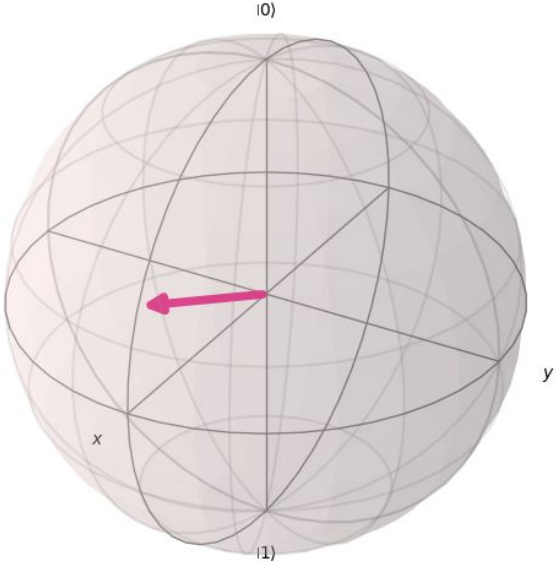
Apply the quantum rotation gate sequence to prepare the qubit state.

tumor_size_m m	x_norm	$\theta = \pi \cdot$ x_norm	$\cos(\theta/2 0\rangle)$	$\sin(\theta/2 1\rangle)$	Encoded State $ \psi\rangle$
25	$(25-17)/23 =$ $8/23 =$ 0.3478	$\pi \cdot$ $0.3478 =$ 62.6	$\cos(62.6/2)$ $= 0.8544$	$\sin(62.6/2)$ $= 0.5195$	$ \psi\rangle =$ $0.854 0\rangle +$ $0.519 1\rangle$



Age	x_norm	$\theta = \pi \cdot x_norm$	$\cos(\theta/2 0\rangle)$	$\sin(\theta/2 1\rangle)$	Encoded State $ \psi\rangle$
50.42	$(50.42 - 39.84) / (68.68 - 39.84) = 0.36685$	$\pi \cdot 0.36685 = 1.1525 = 66.03^\circ$	0.8385	0.5449	$ \psi\rangle \approx 0.8385 0\rangle + 0.5449 1\rangle$

Age 50.42 Angle Encoded: $\theta = 66.03^\circ$



1. Encoding & Representation

Role in A-QLSTM-A

Encoding is the critical step that converts raw, classical clinical features into quantum states. After normalization and angle encoding, a feature is transformed into the state of the qubit:

$$|\psi\rangle = 0.854|0\rangle + 0.519|1\rangle$$

These encoded qubits flow directly into the quantum LSTM network where gates, memory cells, and attention layers process them for TNBC classification.

The Translation Principle

INPUT

Numeric Clinical Values Raw patient data such as tumor size and age

TRANSFORMATION

Angle Encoding $R_Y(\theta)$ Mapping continuous variables to Bloch sphere rotations

OUTPUT

Operable Quantum States qubits that the neural network architecture can process